

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 11

DATE : 06-10-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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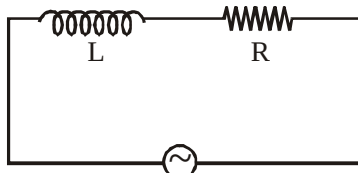
BEWARE OF NEGATIVE MARKING

1. The diagram below shows the lowest four energy levels for an electron in a hypothetical atom. The electron is excited to the -1eV level of the atom and transitions to the lowest energy state by emitting only two photons. Which of the following energies could not belong to either of the photons ?

-1 eV _____
 -3 eV _____
 -7 eV _____
 -12 eV _____

- (1) 2 eV (2) 4 eV
 (3) 5 eV (4) 6 eV

2. In the given figure, the current amplitude is :-



$$V = V_0 \sin \omega t + (V_0/2) \cos \omega t$$

- (1) $\sqrt{\frac{5V_0^2}{4(\omega^2 L^2 + R^2)}}$ (2) $\frac{V_0}{\sqrt{4(\omega^2 L^2 + R^2)}}$
 (3) $\frac{\sqrt{5}V_0}{2\sqrt{\omega^2 L^2 - R^2}}$ (4) $\frac{\sqrt{5}V_0}{2\sqrt{R^2 - \omega^2 L^2}}$

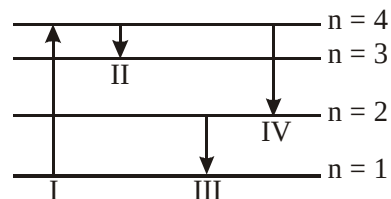
3. A charged capacitor discharges through a resistance R with time constant τ . The two are now placed in series across an AC source of angular frequency $\omega = \frac{1}{\tau}$. The impedance of the circuit will be-

- (1) $\frac{R}{\sqrt{2}}$ (2) R
 (3) $\sqrt{2}R$ (4) $2R$

4. A condenser of capacity $6\text{ }\mu\text{F}$ is fully charged using a 6-volt battery. The battery is removed and a resistanceless 0.2 mH inductor is connected across the condenser. The current which is flowing through the inductor when one-third of the total energy is in the magnetic field of the inductor is :-

- (1) 0.1 A (2) 0.2 A
 (3) 0.4 A (4) 0.6 A

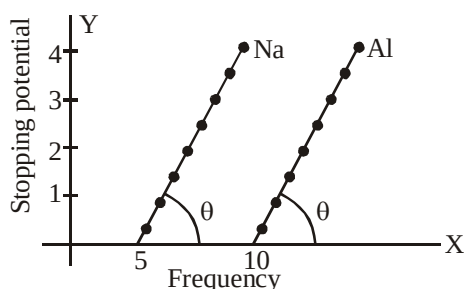
5. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with maximum wavelength?



- (1) III (2) IV
 (3) I (4) II

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6. From the figure describing photoelectric effect we may infer correctly that

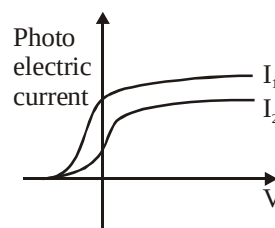


- (1) Na and Al both have the same threshold frequency
(2) Maximum kinetic energy for both the metals depends linearly on the frequency
(3) The change in stopping potentials are different for Na and Al for the same change in frequency
(4) Al is a better photo sensitive material than Na
7. Let F_1 be the frequency of second line of Lyman series and F_2 be the frequency of first line of Balmer series then frequency of first line of Lyman series is given by
(1) $F_1 - F_2$ (2) $F_1 + F_2$
(3) $F_2 - F_1$ (4) $\frac{F_1 F_2}{F_1 + F_2}$
8. Ground state energy of H-atom is -13.6 eV. The energy needed to ionise H-atom from its second excited state is
(1) 1.51 eV (2) 3.4 eV
(3) 13.6 eV (4) 12.1 eV

9. Let K_1 be the maximum kinetic energy of photoelectrons emitted by a light of wavelength λ_1 and K_2 corresponding to λ_2 . If $\lambda_1 = 2\lambda_2$, then :

- (1) $2K_1 = K_2$ (2) $K_1 = 2K_2$
(3) $K_1 < \frac{K_2}{2}$ (4) $K_1 > 2K_2$

10. In an experiment of photoelectric effect, the variation of photoelectric current is plotted against potential difference across emitter and collector plates for different intensities. On the basis of graph choose the INCORRECT option :-



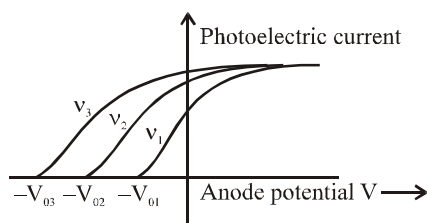
- (1) All emitted photoelectrons do not have same kinetic energy
(2) If frequency of incident light is kept constant, the maximum kinetic energy of emitted photoelectron is independent of intensity.
(3) Maximum value of photoelectric current is independent of intensity of radiation if its frequency is constant.
(4) From the graph, we can conclude radiation intensity $I_1 > I_2$.
11. An α -particle colliding with one of the electrons in a gold atom loses
(1) Most of its momentum
(2) About $\frac{1}{3}$ rd of its momentum
(3) Little of its energy
(4) Most of its energy

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| <p>12. What should be the angular momentum of an electron in Bohr's hydrogen atom whose energy is -0.544 eV?</p> <p>(1) $\frac{h}{\pi}$ (2) $\frac{3h}{2\pi}$ (3) $\frac{5h}{2\pi}$ (4) $\frac{2h}{2\pi}$</p> <p>13. The product of angular speed and tangential speed of electron in n^{th} orbit of hydrogen atom is</p> <p>(1) Directly proportional to n^2
 (2) Directly proportional to n^3
 (3) Inversely proportional to n^4
 (4) Independent of n</p> <p>14. What should be the ratio of minimum to maximum wavelength of radiation emitted by transition of an electron to ground state of Bohr's hydrogen atom?</p> <p>(1) $\frac{3}{4}$ (2) $\frac{1}{4}$ (3) $\frac{1}{8}$ (4) $\frac{3}{8}$</p> <p>15. The ratio of energies of hydrogen atom in its first excited state to third excited state is</p> <p>(1) $\frac{1}{4}$ (2) $\frac{4}{1}$ (3) $\frac{3}{4}$ (4) $\frac{4}{3}$</p> <p>16. Three energy levels L_1, L_2 and L_3 of a hydrogen atom correspond to increasing values of energy i.e., $E_{L_1} < E_{L_2} < E_{L_3}$. If the wavelength corresponding to the transitions L_3 to L_2, L_2 to L_1 and L_3 to L_1 are λ_3, λ_2 and λ_1 respectively then</p> <p>(1) $\lambda_3 = \lambda_1 + \lambda_2$ (2) $\lambda_1 = \frac{\lambda_2 \lambda_3}{\lambda_2 + \lambda_3}$</p> <p>(3) $\lambda_1 = \lambda_2 + \lambda_3$ (4) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$</p> | <p>17. If the wavelength of first member of Lyman series is λ then calculate the wavelength of first member of Pfund series</p> <p>(1) $\frac{675}{11}\lambda$ (2) $\frac{245}{11}\lambda$</p> <p>(3) $\frac{322}{13}\lambda$ (4) $\frac{289}{11}\lambda$</p> <p>18. In Bohr's model of the hydrogen atom, the ratio between the period of revolution of an electron in the orbit of $n = 1$ to the period of revolution of the electron in the orbit $n = 2$ is</p> <p>(1) $2 : 1$ (2) $1 : 2$
 (3) $1 : 4$ (4) $1 : 8$</p> <p>19. How many times do the electron go round the second Bohr orbit in a second ?</p> <p>(1) 2.3×10^{12} (2) 3.2×10^{14}
 (3) 8.3×10^{14} (4) 6.2×10^{14}</p> <p>20. If an electron in hydrogen atom jumps from third orbit to second orbit, the frequency of the emitted radiation is given by (c is speed of light)</p> <p>(1) $\frac{3Rc}{29}$ (2) $\frac{5Rc}{36}$</p> <p>(3) $\frac{7Rc}{36}$ (4) $\frac{8Rc}{31}$</p> <p>21. If longest wavelength of Balmer series of H atom is λ then shortest wavelength of Lyman series will be</p> <p>(1) $\frac{5}{36}\lambda$ (2) $\frac{5}{9}\lambda$ (3) $\frac{36}{5}\lambda$ (4) $\frac{9}{5}\lambda$</p> |
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22. An electron in a hydrogen atom makes a transition such that its kinetic energy increases, then
- (1) The electron may have excited form $n = 2$ to $n = 3$
 - (2) Potential energy of the electron increases
 - (3) Potential energy of the electron decreases
 - (4) Total energy of the electron increases
23. Energy of a quanta of frequency 10^{15} Hz and $h = 6.6 \times 10^{-34}$ J.S will be
- (1) 6.6×10^{-19} J
 - (2) 6.6×10^{-12} J
 - (3) 6.6×10^{-49} J
 - (4) 6.6×10^{-41} J
24. The photoelectric work function
- (1) Is different for different materials
 - (2) Is same for all metals
 - (3) Depends upon frequency of the incident light
 - (4) Depends upon intensity of the incident light
25. In photoelectric effect, the curve between photoelectric current and anode potential V (For different frequencies) is shown in figure, then

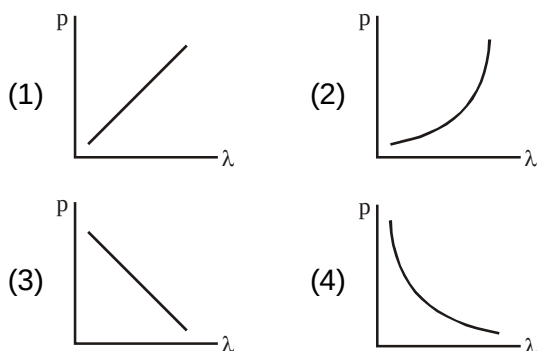


- (1) $v_1 > v_2 > v_3$
- (2) $v_1 < v_2 < v_3$
- (3) $v_1 = v_2 = v_3$
- (4) $v_1 > v_2 < v_3$

26. The number of photo-electrons emitted per second from a metal surface increases when
- (1) The energy of incident photons increases
 - (2) The frequency of incident light increases
 - (3) The wavelength of incident light increases
 - (4) The intensity of incident light increases
27. Wave nature of light cannot explain photoelectric effect because in photoelectric effect, it is seen that :
- (1) For the frequency of light below a certain value, the photoelectric effect does not take place, irrespective of intensity
 - (2) Maximum kinetic energy of ejected electrons is independent of intensity of radiation
 - (3) There is no time lag between the incidence of radiation and emission of electrons
 - (4) All of these
28. When photon of energy 3.8 eV falls on metallic surface of work function 2.8 eV, then the kinetic energy of emitted electrons are :-
- (1) 1 eV
 - (2) 6.6 eV
 - (3) 0 to 1 eV
 - (4) 2.8 eV

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29. A monochromatic point source of light is placed at a distance d from a metal surface. Photoelectrons are ejected at a rate n per second, and with maximum kinetic energy E . If the source is brought nearer to distance $d/2$, the rate and the maximum kinetic energy per photoelectron become nearly
 (1) $2n$ and $2E$ (2) $4n$ and $4E$
 (3) $4n$ and E (4) n and $4E$
30. Photon of 5.5 eV energy falls on the surface of the metal emitting photoelectrons of maximum kinetic energy 4 eV . The stopping potential required for these electrons is:
 (1) 5.5 V (2) 1.5 V
 (3) 9.5 V (4) 4.0 V
31. If in a photoelectric experiment the wavelength of incident radiation is reduced from $6000 \text{ \AA}$ to $4000 \text{ \AA}$ then :
 (1) Stopping potential will decrease
 (2) Stopping potential will increase
 (3) Kinetic energy of emitted electrons will decrease
 (4) The value of work function will decrease
32. Which of the following figures represents the variation of particle momentum (p) and associated de Broglie wavelength (λ) ?



33. Which one of the following statements is not true about de-Broglie waves ?
 (1) All atomic particles in motion have waves character associated with them
 (2) The higher the momentum, the longer is the wave-length
 (3) The faster the particle, the shorter is the wavelength
 (4) For the same velocity, a heavier particle has a shorter wavelength
34. Choose the only correct statement out of the following :
 (1) Only a charged particle in motion is accompanied by matter waves
 (2) Only subatomic particles in motion are accompanied by matter waves
 (3) Any particle in motion, whether charged or uncharged is accompanied by matter waves
 (4) No particle, whether at rest or in motion, is ever accompanied by matter waves
35. The ratio of specific charge q/m of a proton to that of an α -particle is :
 (1) $1 : 4$ (2) $1 : 2$
 (3) $4 : 1$ (4) $2 : 1$
36. What will be the ratio of de-Broglie wavelength of proton and α -particle of same energy :
 (1) $2 : 1$ (2) $1 : 2$
 (3) $4 : 1$ (4) $1 : 4$

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| <p>37. The de-Broglie wavelength of photoelectrons is $1\text{\AA}$. Its accelerating potential is :</p> <p>(1) 150 V (2) 15.3 V
(3) 12.3 V (4) 13.6 V</p> <p>38. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor:</p> <p>(1) $\frac{1}{\sqrt{2}}$ (2) $\sqrt{2}$
(3) $1/2$ (4) 2</p> <p>39. De-broglie wavelength of a body of mass 1 kg moving with velocity of 2000 m/s is :</p> <p>(1) $3.32 \times 10^{-27} \text{\AA}$
(2) $1.5 \times 10^7 \text{\AA}$
(3) $0.55 \times 10^{-22} \text{\AA}$
(4) $2 \times 10^{20} \text{\AA}$</p> <p>40. The de-Broglie wavelength of an electron and the wavelength of a photon are the same. The ratio between the energy of that photon and the momentum of that electron is :-</p> <p>(1) h (2) c
(3) $\frac{1}{h}$ (4) $\frac{1}{c}$</p> <p>41. The de Broglie wavelength of an electron in the n^{th} Bohr orbit is related to the radius R of the orbit as :-</p> <p>(1) $n\lambda = \pi R$
(2) $n\lambda = \frac{3}{2} \pi R$
(3) $n\lambda = 2\pi R$
(4) $n\lambda = 4\pi R$</p> | <p>42. Electric field in an electromagnetic wave is given by $E = 50 \sin \omega (t - x/c) \text{ NC}^{-1}$. Intensity of this wave in Wm^2</p> <p>(1) 50
(2) 1.1×10^8
(3) 3.3
(4) 5.5×10^{-19}</p> <p>43. When light of wavelength 300 nm falls on a photoelectric emitter, photoelectrons are just liberated. For another emitter, light of wavelength 600 nm is just sufficient for liberating photoelectrons. The ratio of the work function of the two emitters is :-</p> <p>(1) 1 : 2 (2) 2 : 1
(3) 4 : 1 (4) 1 : 4</p> <p>44. In an electromagnetic wave, if the amplitude of magnetic field is $3 \times 10^{-10} \text{ T}$, the amplitude of the associated electric field will be _____</p> <p>(1) $9 \times 10^{-2} \text{ Vm}^{-1}$
(2) $3 \times 10^{-10} \text{ Vm}^{-1}$
(3) $3 \times 10^{-2} \text{ Vm}^{-1}$
(4) $1 \times 10^{-18} \text{ Vm}^{-1}$</p> <p>45. The electric and magnetic field of an electromagnetic wave are</p> <p>(1) in phase and perpendicular to each other
(2) in phase and parallel to each other
(3) in opposite phase and perpendicular to each other
(4) in opposite phase and parallel to each other</p> |
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| <p>46. Covalency of Carbon in CCl_4 will be
 (1) 1 (2) 2
 (3) 3 (4) 4</p> <p>47. Two hybrid orbitals have a bond angle of 120°. The percentage of p character in the hybrid orbital is nearly:-
 (1) 25% (2) 33%
 (3) 50% (4) 66%</p> <p>48. The energy of BMO is less than combining atomic orbital because :-
 (1) Molecular orbital are more asymmetrical than atomic orbital.
 (2) Molecular orbitals are near to nucleus than atomic orbitals.
 (3) Probability of finding the electron in BMO is less than atomic orbital.
 (4) Probability of finding the electron in BMO is more than atomic orbital.</p> <p>49. Which of the following statement is correct :-
 As the S-character of a hybrid orbital decreases
 (1) The bond angle decreases
 (2) The bond strength increases
 (3) The bond length decreases
 (4) Size of orbital decreases</p> <p>50. Which of the following set do not have sp^3d hybridization
 (1) PF_4^- & BrF_3
 (2) ICl_2^+ & SF_4
 (3) XeF_2 & I_3^-
 (4) AsF_4^- & SCl_4</p> | <p>51. Correct order of extent of overlapping is :-
 (1) $2s - 2s > 2p - 2p > 2s - 2p$
 (2) $1s - 1s > 2p - 2p > 2s - 2p$
 (3) $1s - 1s < 2s - 2s < 3s - 3s$
 (4) $3s - 3s > 3s - 3p > 3p - 3p$</p> <p>52. Shape of molecule having 4-bond pair and one lone pair is :-
 (1) Trigonal bipyramidal
 (2) T-Shape
 (3) See-saw
 (4) Square planar</p> <p>53. Intramolecular hydrogen bonding is not present in :-
 (1) p-fluoro phenol
 (2) Chloral hydrate
 (3) o-nitro phenol
 (4) o-fluoro phenol</p> <p>54. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
 (i) Cu metal is extracted from its sulphide ore by reduction of Cu_2O with FeS.
 (ii) An ore of Tin containing FeWO_4 is concentrated by magnetic separation method.
 (iii) Auto reduction process is used in the extraction of Cu & Hg.
 (iv) Cassiterite and Rutile are oxide ores of the metals.</p> <p>(1) TFFT (2) TTFT
 (3) FTTT (4) FFFT</p> |
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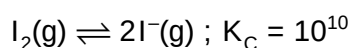
55. Which is the correct decreasing order of covalent nature :-

- (1) $\text{SiCl}_4 > \text{AlCl}_3 > \text{ZnCl}_2 > \text{CaCl}_2 > \text{MgCl}_2$
- (2) $\text{SiCl}_4 > \text{AlCl}_3 > \text{MgCl}_2 > \text{CaCl}_2 > \text{ZnCl}_2$
- (3) $\text{ZnCl}_2 > \text{SiCl}_4 > \text{AlCl}_3 > \text{MgCl}_2 > \text{CaCl}_2$
- (4) $\text{SiCl}_4 > \text{AlCl}_3 > \text{ZnCl}_2 > \text{MgCl}_2 > \text{CaCl}_2$

56. If the forward rate constant of reversible reaction is 0.16 and backward rate constant is 4×10^4 , then the equilibrium constant will be.

- (1) 2.25×10^{-6}
- (2) 2.5×10^{-5}
- (3) 4×10^{-6}
- (4) 4×10^{-5}

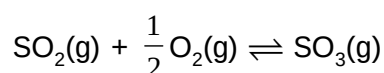
57. In the dissociation of $\text{I}_2(\text{g})$ at 1000 K in a container of 1 litre :-



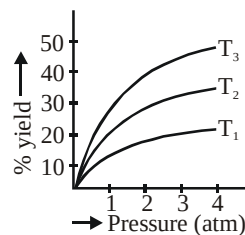
Select the correct relation:-

- (1) $[\text{I}_2] > [\text{I}^-]$
- (2) $[\text{I}_2] < [\text{I}^-]$
- (3) $[\text{I}_2] = [\text{I}^-]$
- (4) $[\text{I}_2] = \frac{1}{2} [\text{I}^-]$

58. The preparation of $\text{SO}_3(\text{g})$ by reaction



is an exothermic reaction. If the preparation follows the following temperature-pressure relationship for its % yield, then for temperatures T_1, T_2 & T_3 . The correct option is



- (1) $T_3 > T_2 > T_1$
- (2) $T_1 > T_2 > T_3$
- (3) $T_1 = T_2 = T_3$
- (4) Nothing could be predicted about temperature through given information.

59. The pH of solution is increased from 3 to 6. Its H^+ ion conc. will be :-

- (1) Reduced to half
- (2) Doubled
- (3) Reduced by 1000 times
- (4) Increased by 1000 times

60. FeCl_3 solution is acidic due to hydrolysis of :-

- (1) FeCl_3
- (2) Cl^-
- (3) Fe^{+3}
- (4) None of these

61. A solution of x moles of sucrose in 100 grams of water freezes at -0.2°C . Calculate the amount of ice separate out when it is cooled upto -0.25°C :-

- (1) 18 grams
- (2) 20 grams
- (3) 25 grams
- (4) 23 grams

62. The molarity of a solution made by mixing 50 ml of conc. H_2SO_4 (36 N) with 50 ml of water, is :

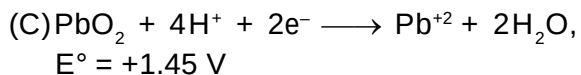
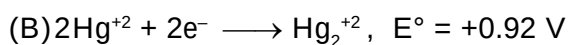
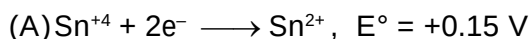
- (1) 36 M
- (2) 18 M
- (3) 9 M
- (4) 6 M

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63. Vapour pressure of CCl_4 at 25°C is 143 mm Hg. 0.5 gm of a non-volatile solute (mol. wt. 65) is dissolved in 100 ml of CCl_4 . Find the vapour pressure of the solution.
(Density of $\text{CCl}_4 = 1.58 \text{ gm/cm}^3$)
(1) 141.93 mm
(2) 94.39 mm
(3) 199.34 mm
(4) 143.99 mm
64. 150ml of $\text{C}_2\text{H}_5\text{OH}$ (density = 0.78 gm/ml) is diluted to one liter by adding water; molality of the solution is –
(1) 2.54
(2) 11.7
(3) 2.99
(4) 29.9
65. pH of a 0.1 M monobasic acid is measured to be 2. Its osmotic pressure at a given temperature T K is :-
(1) 0.1 RT
(2) 0.11 RT
(3) 1.1 RT
(4) 0.01 RT
66. A solution containing 0.1 g of a non-volatile organic substance P (molecular mass 100) in 100g of benzene raises the boiling point of benzene by 0.2°C , while a solution containing 0.1 g of another non-volatile substance Q in the same amount of benzene raises the boiling point of benzene by 0.4°C . What is the ratio of molecular masses of P and Q ?
(1) 1 : 2
(2) 2 : 1
(3) 1 : 4
(4) 4 : 1
67. Two solutions of a substance (non-electrolyte) are mixed in the following manner 480 mL of 1.5M of first solution with 520 mL of 1.2 M of second solution. The molarity of final solution is :-
(1) 1.20 M
(2) 1.50 M
(3) 1.344 M
(4) 2.70 M
68. The emf of the cell
 $\text{Zn} \mid \text{Zn}^{2+} (0.01 \text{ M}) \parallel \text{Fe}^{2+} (0.001 \text{ M}) \mid \text{Fe}$
at 298 K is 0.2905 then the value of equilibrium constant for the cell reaction is :-
(1) $e^{\frac{0.32}{0.0295}}$
(2) $10^{\frac{0.32}{0.0295}}$
(3) $10^{\frac{0.26}{0.0295}}$
(4) $10^{\frac{0.32}{0.0591}}$
69. Calculate the cell potential of the following concentration cell :
 $\text{Pt} \mid \text{Cl}_2 (0.4 \text{ bar}) \mid \text{Cl}^- (0.1 \text{ M}) \parallel \text{Cl}^- (0.01 \text{ M}) \mid \text{Cl}_2 (0.2 \text{ bar}) \mid \text{Pt}$
(1) 0.051 V
(2) – 0.051 V
(3) 0.102 V
(4) 0.0255 V
70. Consider the following four electrodes,
P = $\text{Cu}^{2+} (0.0001 \text{ M}) / \text{Cu(s)}$
Q = $\text{Cu}^{2+} (0.1 \text{ M}) / \text{Cu(s)}$
R = $\text{Cu}^{2+} (0.01 \text{ M}) / \text{Cu(s)}$
S = $\text{Cu}^{2+} (0.001 \text{ M}) / \text{Cu(s)}$
If the standard reduction potential of $\text{Cu}^{2+} / \text{Cu}$ is 0.34 V, the reduction potential in volts of the above electrodes follow the order :-
(1) $Q > R > S > P$
(2) $P > S > R > Q$
(3) $R > S > Q > P$
(4) $P > Q > R > S$

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71. Standard electrode potential of some half cell reactions are given below -



for these which statement is correct -

(1) Pb^{+2} is more powerful reducing agent than Sn^{+2}

(2) Sn^{+4} is more powerful oxidising agent than Pb^{+4}

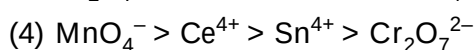
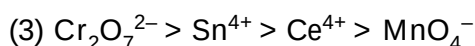
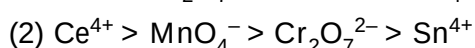
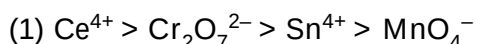
(3) Sn^{+2} is more powerful reducing agent than Hg_2^{+2}

(4) Hg^{+2} is more powerful oxidising agent than Pb^{+4}

72. The standard reduction potentials E° of the following systems are

System	E° (Volts)
(i) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51
(ii) $\text{Sn}^{4+} + 2\text{e}^- \rightarrow \text{Sn}^{2+}$	0.15
(iii) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.33
(iv) $\text{Ce}^{4+} + \text{e}^- \rightarrow \text{Ce}^{3+}$	1.61

The oxidising power of the various species decreases in the order :-



73. In order to refine "copper" it is melted in a furnace and it is stirred with green logs of wood. The purpose is :-

(1) to expel the dissolved gases in blister copper

(2) to reduce the metallic oxide impurities with hydrocarbon gases liberated from the wood

(3) to bring the impurities to surface and oxidise them

(4) to increase the carbon content in copper

74. Read the following statements :-

(I) Al has greater affinity than that of Cr, for oxygen

(II) Al can be used for reduction of Fe_2O_3

(III) Carbon is used for reduction of copper oxide (CuO)

(IV) SnO_2 can be reduced by carbon

Choose the correct set of statement(s)

(1) I, III

(2) I, II, IV

(3) I, II, III

(4) All of these

75. Which of the following is incorrect ?

(1) Leaching is a method to remove quantitatively Fe_2O_3 from red bauxite

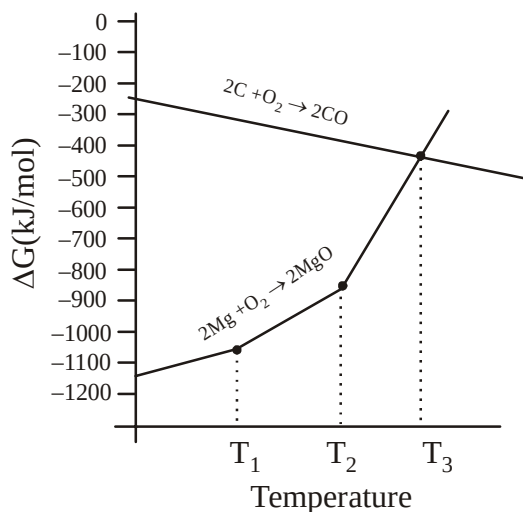
(2) Metallurgy of gold involves pyrometallurgical process form soluble complex formation

(3) Metallurgy of silver involves hydrometallurgical process form soluble complex formation

(4) Self reduction is not possible during metallurgy of Mg

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76. Select the incorrect statements about the plot is / are:



- (1) T_1 and T_2 are melting point & boiling point of Mg respectively.
- (2) T_1 and T_2 are melting point & boiling point of MgO respectively.
- (3) Reduction of MgO by coke is possible above T_3
- (4) Mg can be extracted from gaseous products by rapid cooling.

77. Which of the following statement is not correct regarding calcination ?

- (1) Impurities are removed in the form of elemental vapours
- (2) Carbonate ores convert into their oxides
- (3) Temperature of the process is maintained below the melting point of the mixture
- (4) Lower oxidation state oxides are oxidised further

78. Which of the following is the principal reducing agent for the reduction of Fe_2O_3 :

- (1) CO
- (2) C
- (3) CO_2
- (4) None of these

79. Which of the following is not correctly matched :-

- (1) Removal of oxide impurity from impure Cu $\Rightarrow$ Poling process
- (2) Removal of Pb impurity from impure Ag $\Rightarrow$ Cupellation
- (3) Obtaining wrought iron from cast iron – Bessemerisation
- (4) Refining of Nickel – Mond's process

80. The method of zone refining of metals is based on the principle of :-

- (1) Greater mobility of the pure metal than that of impurity
- (2) higher melting point of the impurity than that of the pure metal
- (3) Greater noble character of the solid metal than that of the impurity
- (4) Greater solubility of the impurity in the molten state than in the solid

81. The reduction of an oxide by aluminium is called :-

- (1) Baeyer's process
- (2) Goldschmidt's aluminothermite process
- (3) Hall's process
- (4) van Arkel process

82. Which of the following sulphide ore is not concentrated by froth-floatation process ?

- (1) Galena
- (2) Zinc blend
- (3) Argentite
- (4) Cinnabar

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- | | |
|---|---|
| <p>83. Just after the slow crystallization of a solvent containing non volatile solute the concentration of solution :-</p> <p>(1) Increases</p> <p>(2) Decreases</p> <p>(3) Constant</p> <p>(4) None</p> <p>84. The van't Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is :-</p> <p>(1) 10</p> <p>(2) 4</p> <p>(3) 5</p> <p>(4) 0.25</p> <p>85. Binary liquid solutions which exhibit negative deviations from Raoult's law boil at temperature.... than the expected value :</p> <p>(1) lower</p> <p>(2) higher</p> <p>(3) same</p> <p>(4) cannot be said</p> <p>86. Froth floatation process for the concentration of sulphide ores is an illustration of the practical application of -</p> <p>(1) Adsorption</p> <p>(2) Absorption</p> <p>(3) Sedimentation</p> <p>(4) Coagulation</p> | <p>87. Which of the following process is not a physical process of separation</p> <p>(1) Levigation</p> <p>(2) Magnetic separation</p> <p>(3) Leaching</p> <p>(4) Froth floatation</p> <p>88. Among the following groups of oxides, the group containing oxides that cannot be reduced by C to give the respective metal is</p> <p>(1) CaO and K_2O</p> <p>(2) Fe_2O_3 and ZnO</p> <p>(3) Cu_2O and SnO_2</p> <p>(4) PbO and Pb_3O_4</p> <p>89. In Goldschmidt aluminothermic process, thermite mixture contains :</p> <p>(1) 3 parts Fe_2O_3 and 2 parts Al</p> <p>(2) 3 parts Al_2O_3 and 4 parts Al</p> <p>(3) 1 part Fe_2O_3 and 12 part Al</p> <p>(4) 3 parts Fe_2O_3 and 1 part Al</p> <p>90. Which of the following metals are obtained by auto reduction method:</p> <p style="text-align: center;">Pb, Mn, Cu, Cr, Fe, Al.</p> <p>(1) Cu, Fe</p> <p>(2) Cu, Pb, Mn</p> <p>(3) Mn, Cr, Pb</p> <p>(4) Pb, Cu</p> |
|---|---|

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|--|--|
| <p>91. Photosynthesis is :
 (1) Bio-chemical process
 (2) Physico-chemical process
 (3) Thermo-chemical process
 (4) Electro-chemical process</p> <p>92. Photosynthesis is important due to :
 (1) It is the primary source of all food
 (2) It is also release of oxygen
 (3) Both (1) and (2)
 (4) It is the source of ATP of cell</p> <p>93. What are required for photosynthesis to occur :
 (1) Light (2) CO_2
 (3) e^- donor (4) All</p> <p>94. CO_2 is required for photosynthesis, we can demonstrated by :
 (1) Priestley experiment
 (2) Four leaf experiment
 (3) T. W. Engelmann experiment
 (4) Half leaf experiment</p> <p>95. In photosynthesis, source of O_2 is water, this was proved using :
 (1) X-ray differection technique
 (2) Radio-isotope technique
 (3) Chromatography
 (4) None of these</p> <p>96. In cactus, photosynthesis does take place in :-
 (1) The spines (2) The green stem
 (3) The green root (4) All of these</p> <p>97. What type of movement exhibited by chloroplast in mesophyll cells to the incident light :
 (1) Chemotropic (2) Phototropic
 (3) Phototactic (4) Photonastic</p> <p>98. Which is the most abundant plant pigment in world :
 (1) Chlorophyll-a + b
 (2) Chlorophyll-b + c
 (3) Chlorophyll-c + d
 (4) Chlorophyll-d + a</p> | <p>99. Determine the wavelength in which chlorophyll-a show maximum absorption :
 (1) Red - Blue (2) Red - Far red
 (3) Blue - Red (4) Green - Red</p> <p>100. RUBISCO enzyme is also called as _____.
 (1) carboxytetra mutase
 (2) carboxydimutase
 (3) carboxytrimutase
 (4) carboxyunimutase</p> <p>101. In the electron transport system of chloroplast, a molecule that has a greater redox potential behave as
 (1) Electron donor
 (2) Electron acceptor
 (3) Both (1) and (2)
 (4) Electron user</p> <p>102. The first discovered restriction endonuclease enzyme is
 (1) Hind II (2) Hind I
 (3) EcoRI (4) BamHI</p> <p>103. In EcoRI, the letter 'R' is derived from :
 (1) The name of strain
 (2) The name of species
 (3) The name of genus
 (4) None of these</p> <p>104. Which enzyme cut DNA :
 (1) Restriction endonuclease
 (2) Exonuclease
 (3) Endonuclease
 (4) All these</p> <p>105. Since DNA fragments are negatively charged molecules can be separated by forcing them to move under electric field towards :
 (1) Cathode (2) Anode
 (3) Both (4) Not conform</p> <p>106. Which is wrong option :
 (1) We can not see pure DNA fragments in the visible light and without staining.
 (2) Stained DNA band only visible under UV-exposure.
 (3) Elution step includes cut out the coloured DNA bands from agarose gel & extraction from the gel piece.
 (4) Agarose is a natural polymer that shows high affinity with DNA.</p> |
|--|--|

107. During running of gel electrophoresis what will the location of DNA fragments in agarose gel :
- (1) Larger → towards cathode and near cathode Smaller → towards anode and near anode
 - (2) Larger → towards anode and near anode Smaller → far away from cathode and near anode
 - (3) Larger → towards anode and near cathode Smaller → towards anode and near anode
 - (4) Larger → far away from anode and near cathode Smaller → towards cathode and near anode
108. Which is not a tool for genetic engineering :
- (1) DNA polymerase
 - (2) DNA ligase
 - (3) Plastid
 - (4) Molecular scissor
109. Which of the following feature is required to facilitate cloning into a vector :
- (1) Ori - site
 - (2) Selectable marker
 - (3) Cloning site
 - (4) All are required
110. Which of the following cuts at specific positions with in the DNA :
- (1) Some nucleases enzymes
 - (2) Exonucleases enzymes
 - (3) Endonucleases enzymes
 - (4) Exoendonucleases
111. Sexual reproduction is better than asexual reproduction because it permitt variations in genetic material which is better for organisms or population but there is problem that with desirable characters some undesirable characters also enter in offspring, which is solve by using the :
- (1) Biotechnology
 - (2) Genetic engineering
 - (3) Tissue culturing
 - (4) More than one option is correct
112. Group of letters that form the same words when read both forward and backward like
- 5' – GAATTC – 3'
- 3' – CTTAAG – 5'
- Which of the following is incorrect for such sequences :
- (1) Such sequences known as pallindrome site which are recogination sites for different restriction enzymes
 - (2) These sequences of base pairs at DNA strand remain same when orientation of reading is kept same
 - (3) For specific Palindrome specific restriction enzyme have recognition site
 - (4) Different restriction enzymes cut the same pallindrome site in vector DNA and foreign DNA during formation of r-DNA
113. _____ ions help in photolysis of water.
- (1) Mn⁺⁺
 - (2) Mg⁺⁺
 - (3) Cl⁻
 - (4) (1) and (3) both
114. In recombinant DNA technique vector is referred as :
- (1) Cosmids that can cut DNA at specific base sequence
 - (2) Plasmids that can join different DNA fragments
 - (3) Cosmids that can degrade harmful proteins
 - (4) Plasmids that can transfer foreign DNA into a living cell
115. Which of the following is not include under new biotechnology :
- (1) Correcting a defective gene through gene therapy
 - (2) Gene manipulation in genetic material of any organism
 - (3) Developing a DNA vaccine
 - (4) Production of citric acid by *Aspergillus niger*.
116. Human insulin is produced by the :
- (1) E. coli
 - (2) Salmonella
 - (3) Yeast
 - (4) Tobacco

117. The southern blotting technique depends on -
- (1) similarities between the sequences of probe DNA and experimental DNA
 - (2) molecular mass of proteins
 - (3) amino acid sequence of a protein
 - (4) All
118. DNA that is obtained using processed mRNA molecules as a template
- (1) rDNA
 - (2) mDNA
 - (3) cDNA
 - (4) plasmid DNA
119. Consider the given statements -
- (A) Less than 900 restriction enzymes have been isolated from over 230 strains of bacteria.
 - (B) Transformants will grow on specific antibiotic containing agar plate while non-transformants are not grow at this agar plate.
 - (C) PCR is used to synthesize multiple copies of gene of interest in vitro type.
 - (D) Most commonly using bioreactors are stirring type in biotechnical product industries.
 - (E) T-DNA of *agrobacterium tumifaciens* bacteria which transform normal plant cells into tumor are present at nucleoid DNA.
- How many are incorrect statements :
- (1) 5
 - (2) 3
 - (3) 2
 - (4) 1
120. The term 'B' and 'R' in the name P^{BR322} represents -
- (1) Name of the scientists
 - (2) Name of the restriction endonuclease enzyme
 - (3) Name of the cities
 - (4) Name of the used bacterial strain
121. Transformation is a process through which a piece of DNA is introduced in a hostA..... while transfection is a procedure through which a piece of DNA is introduced in aB..... host cell
Here A and B are correctly :
- | A | B |
|---------------------|-----------------|
| (1) Bacteria cell | E.Coli |
| (2) Eukaryotic cell | Plant cell |
| (3) Bacteria cell | Eukaryotic cell |
| (4) Eukaryotic cell | Bacteria cell |
122. In relation to a fungi cell for isolation of genetic material which of the following is required :
- (1) Lysozyme enzyme treatment
 - (2) Cellulose enzyme treatment
 - (3) DNA ligase enzyme treatement
 - (4) Chitinase enzyme treatement
123. How is Bt—toxin known to kill the target insects in protection of cotton plants -
- (1) Midgut cells lysis and swelling
 - (2) Paralysis
 - (3) Formation of abnormal proteins
 - (4) Brain death
124. When foregin DNA is inserted in coding sequence of β -galactosidase enzyme in plasmid DNA then what is happen :
- (1) Gene expression for β -galactosidase take place
 - (2) Synthesis of β -galactosidase enzyme become present by recombinant DNA
 - (3) Blue colour colonies occured in culture medium by recombinants
 - (4) Gene expression for β -galactosidase is not take place
125. To take up the recombinant DNA, host bacterium is treated with :
- (1) Heat shock at 42°C
 - (2) Specific concentration of a divalent cation like Ca^{+2}
 - (3) Retroviruses are placed in contact with host bacterium
 - (4) (1) and (2) both

126. Thermostable DNA polymerase which used in PCR is isolated fromA..... which works atB..... A and B are correctly :

A

B

- | | |
|------------------------------|------------|
| (1) A bacterium | Below 40°C |
| (2) <i>Thermus aquaticus</i> | Above 70°C |
| (3) A Fungus | Above 70°C |
| (4) A bacteriophage | Above 90°C |

127. Recombinant proteins are generally produce in :

- (1) Heterologous host
- (2) Homologous host
- (3) Any one of both
- (4) None of these

128. In PBR³²² cloning vector for which of the following restriction endonuclease enzyme restriction site is not present :

- | | |
|-----------|--------------|
| (1) Pvu I | (2) Hind III |
| (3) Sal I | (4) Hae III |

129. Which of the following is not attack at DNA strand :

- | | |
|------------|---------------|
| (1) Eco RI | (2) Hind III |
| (3) RNase | (4) Nucleases |

130. Correctly matched List I with List II

List - I

List - II

- | | |
|---------------------------|---|
| A. Selectable marker gene | (i) By treatment of ice and heat shock. |
| B. Recombinant protein | (ii) Large volumes of culture can be processed. |
| C. Bioreactor | (iii) Tetracycline resistant gene |
| D. Competent host | (iv) Humulin |
| | (v) Insertional inactivation take place |

Correct match is :

- | A | B | C | D |
|------------|-------|--------|----|
| (1) i | ii, v | iii | iv |
| (2) iii, v | iv | ii | i |
| (3) iii | iv, v | i | ii |
| (4) v | iv | i, iii | ii |

131. Besides better aeration and mixing properties what is other advantage of stirred tank bioreactors have over shake flasks :

- (1) Other microbes become die due to stirring
- (2) Create foam control system
- (3) Increase nourishment for producing proteins
- (4) None of these.

132. A : Large scale production of the products is carried out in bioreactors in r-DNA technology.

R : Bioreactors provide optimum growth conditions.

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is NOT the correct explanation of A.
- (3) A is true and R is false.
- (4) A is false and R is true/false.

133. Golden rice is rich source of :

- (1) Proteins
- (2) Vitamin-C
- (3) Vitamin-A
- (4) Protein and Vitamin-C

134. A : Plasmid is extra chromosomal circular DNA of bacteria.

R : Plasmid is ss-DNA found in some bacteria.

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is NOT the correct explanation of A.
- (3) A is true and R is false.
- (4) A is false and R is true/false.

135. Annealing step of PCR is take place in presence of :

- (1) Primers, Taq DNA polymerase enzyme
- (2) Primers below to 80°C
- (3) Primers above to 80°C
- (4) Taq DNA polymerase enzyme

136. Few statements are given below about origin of universe. Choose the correct answer :-
- (1) Universe is around 20 million years old
 - (2) The big bang theory explains about origin of life on earth
 - (3) Universe comprised of galaxies
 - (4) The gases decondensed under the gravitation and formed galaxies of the present day universe
137. Which of the following can contribute to genetic drift?
- (1) Small population size
 - (2) Bottleneck effect
 - (3) Founder effect
 - (4) All of the above
138. Unit of evolution is:
- (1) Population
 - (2) Gene
 - (3) Species
 - (4) Individual
139. Read the sentences carefully about fossils. Choose the correct option which defines the word "fossil" better :-
- (1) Fossils are soft part of life forms found in rocks
 - (2) Fossils are hard part of life forms found in water
 - (3) Fossils are hard part of life forms found in igneous rock
 - (4) Fossils are remains of hard part of life found in rocks
140. Unit of natural selection is:
- (1) Individual
 - (2) Population
 - (3) Species
 - (4) Gene
141. According to Hugo deVries mutations are:
- (1) Large
 - (2) Random
 - (3) Directionless
 - (4) All of the above
142. Lederberg's replica plate experiment proves that:
- (1) Pre existed advantageous mutations provide the selective advantage to individuals in the changed environment
 - (2) Mutations appear after change in environment and provide selective advantage
 - (3) Both conclusions are correct
 - (4) Change in environment doesn't affect the populations
143. The original variety of Darwin finches was :-
- (1) Black and insect eater
 - (2) Grey and seed eater
 - (3) Black and seed eater
 - (4) Grey and insect eater
144. Who had given the theory of evolution of species by natural selection :
- (1) Mendel
 - (2) Darwin
 - (3) Morgan
 - (4) Dalton
145. The Big Bang explosion was :-
- (1) Singular and imaginable in physical terms
 - (2) Multiple and unimaginable in physical terms
 - (3) Singular but can't imaginable in physical terms
 - (4) Multiple but imaginable in physical terms
146. According to abiogenesis, the life appears on earth from :-
- (1) Special method
 - (2) Universe
 - (3) Rotting and decaying material
 - (4) Pre existing life
147. The possibilities of origin of life were established in the water because :-
- (1) The water was very hot
 - (2) Source of energy was sun
 - (3) Components of protoplasm were present in oceanic water
 - (4) Oxidising atmosphere

148. Long and thick body hairs present on humans are an example of :-
 (1) Vestigial organ
 (2) Reversion
 (3) Connecting link
 (4) Functional organ
149. The connecting link between fishes and amphibia is:-
 (1) Peripatus (2) Protopterus
 (3) Platypus (4) Seymauria
150. Carrot and ginger are an example of :-
 (1) Homologous organ
 (2) Analogous organ
 (3) Divergent evolution
 (4) Both (1) and (3)
151. The evolution was termed as 'RNA World' on the basis of specific property of RNA molecule. This property was:
 (1) Enzymatic property
 (2) Duplication property
 (3) Semi conservative duplication property
 (4) Acting as a template for protein synthesis
152. Which type of plants were fell to form coal deposits ?
 (1) Angiosperms (2) Gymnosperms
 (3) Pteridophytes (4) Bryophytes
153. Dinosaurs were originated in which period ?
 (1) Mesozoic (2) Triassic
 (3) Jurassic (4) Cretaceous
154. Which of the following structure's is analogous to the wing of a bird ?
 (1) Dorsal fin of a shark
 (2) Wing of butterfly
 (3) Hind limb of rabbit
 (4) Flipper of whale
155. Evolution of different species in a given area starting from a point and spreading to other geographical areas is known as :-
 (1) Migration
 (2) Convergent evolution
 (3) Adaptive radiation
 (4) Natural selection
156. Which theory states that life is immutable and has not changed ever since its origin?
 (1) Cosmozoic theory
 (2) Special creation theory
 (3) Biogenesis theory
 (4) Abiogenesis theory
157. Which was not a consequence of liberation of oxygen by cyanobacteria on earth?
 (1) Atmosphere became oxidizing
 (2) Formation of ozone layer
 (3) Aerobic mode of respiration started
 (4) Chances of further chemical evolution increased
158. Which of the following is mismatched pair of animals and its related information :
 (1) *Latimeria* (Coelacanth) - A living fossil
 (2) *Archaeopteryx* - A missing link
 (3) *Tyrannosaurus rex* - A large herbivorous dinosaur
 (4) *Pteranodon* - A flying donosaur
159. Select a placental mammal which shows the parallelism with *Numbat*?
 (1) Bobcat (2) Ant eater
 (3) Lemur (4) Flying squirrel
160. Which of the following is represented between Lemur and spotted cuscus?
 (1) Parallel evolution
 (2) Divergent evolution
 (3) Adaptive radiation
 (4) Natural selection
161. Which of the following is/are vestigeal structures in humans?
 (1) Caudal vertebrae
 (2) Wisdom tooth
 (3) Vermiform appendix
 (4) All of these
162. The connecting link between annelids and arthropods is
 (1) *Protopterus*
 (2) *Platypus*
 (3) *Neopilina*
 (4) *Peripatus*

163. Analysis of meteorite contents tells us that:

- (1) Life came on earth due to meteorite hits
- (2) Similar processes are occurring elsewhere in space that had occurred on earth
- (3) All stars originated at same time
- (4) Water is essential for origin of life

164. Which of the following is the missing link between birds and reptiles?

- (1) *Seymouria* (2) *Lycaenops*
- (3) *Ichthyostega* (4) *Archaeopteryx*

165. The strongest evidence for change over a long period of time comes from

- (1) Analogous organs
- (2) Paleontological evidence
- (3) Biogeographical
- (4) Embryo studies

166. Select the duct of gall bladder :

- (1) Hepatic duct
- (2) Stenson's duct
- (3) Common hepato-pancreatic duct
- (4) Cystic duct

167. Insulin is secreted by which pancreatic cells ?

- (1) Alpha cells (2) Beta cells
- (3) Delta cells (4) Gamma cells

168. Largest gland of body ?

- (1) Pancreas (2) Duodenum
- (3) Liver (4) Throid

169. The structure which prevents entry of food into wind pipe during swallowing in mammals ?

- (1) Larynx (2) Glottis
- (3) Epiglottis (4) Pharynx

170. Glycogen is stored in :

- (1) Blood (2) Liver
- (3) Lungs (4) Kidney

171. Ptyalin is secreted by ?

- (1) Stomach
- (2) Salivary gland
- (3) Pancreas
- (4) Bile

172. Castale's intrinsic factor helps in absorption of and secreted by cells of gastric glands :

- (1) Cyanocobalamine — Chief cells
- (2) Vitamin B₁₂ — Oxyntic cells
- (3) Vitamin B₁₂ — Parietal cells
- (4) Any two of these

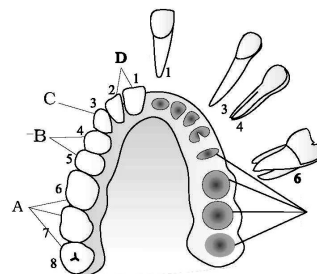
173. The hardest constituent of the tooth is :

- (1) Enamel (2) Dentine
- (3) Bone (4) Pulp

174. Select an incorrect statement about the lipid absorption

- (1) Micelles are formed in cavity of small intestine.
- (2) Chylomicrons are synthesized in intestinal cells
- (3) Bile salts help in micelle formation
- (4) Chylomicrons are absorbed mainly in blood.

175. Identify A, B, C & D in the given diagram of human teeth



- (1) A = Incisor, B = Molar, C = Canine, D = Premolar
- (2) A = Molar, B = Premolar, C = Canine, D = Incisor
- (3) A = Premolar, B = Molar, C = Incisor, D = Canine
- (4) A = Canine, B = Incisor, C = Premolar, D = Molar

176. Emulsification of fat is carried out by:

- (1) Bile pigments
- (2) Bile salts
- (3) HCl
- (4) Pancreatic juice

177. The chemical process of digestion is initiated by the action of : (1) Salivary Amylase (2) Maltase (3) Peptidase (4) Trypsin	179. Which one of the following is not a cell of gastric glands : (1) Mucus neck cells (2) Paneth cells (3) Peptic cells (4) Parietal cells
178. Which cells of mucous layer of stomach secretes pepsinogen : (1) Chief cells (2) Goblet cells (3) Parietal cells (4) Oxyntic cells	180. Bile is formed in : (1) Gall bladder (2) Liver (3) Spleen (4) Blood

SPACE FOR ROUGH WORK

Result : NEET (UG) 2019

ALLEN है... तो मुमकिन है

AIR



8 in Top 10 All India Ranks



61 in Top 100 All India Ranks

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Power of **ALLEN**



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